Problem Statement 2: Modified Program

Dynamic Road Junction Graph with Insertion, Deletion, and Neighbour Queries

A road network consists of several junctions connected by roads. The network is represented using an undirected graph, where junctions are vertices and roads are edges. In a real-world road network, new junctions may be added, and existing roads may be removed. Therefore, the graph must support dynamic modifications.

The objective is to develop a menu-driven C++ program that represents the road network using an adjacency list and allows the user to perform different operations dynamically. The program initially accepts the number of junctions and roads and stores their connections.

The program allows the user to insert a new vertex along with its connections to existing junctions. It also allows the user to remove an edge between two junctions. Additionally, the program calculates the degree of a selected junction and displays its direct neighbours, which are the junctions directly connected to it by roads.

After every insertion or deletion, the adjacency list is updated to maintain the correct graph structure. The program continues executing until the user selects the exit option.

Main Operations:

1. Display the current graph.
2. Insert a new vertex with its associated edges.
3. Remove an existing edge between two junctions.
4. Calculate the degree of a selected vertex.
5. Display the direct neighbours of a selected vertex.
6. Display the updated graph after modifications.
7. Exit the program.

Objective: To efficiently manage a changing road network and understand graph operations such as vertex insertion, edge deletion, degree calculation, and neighbour identification.